ClaimSense AI · Checkpoint 1 Submission

FDE Spine Capstone · Milestones 1 & 2 Checkpoint

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Tech Stack: AWS + OpenAI Agents SDK + Python 3.12 (FastAPI) + Docker + Terraform
Submission Type: Checkpoint - 1 (Milestones 1 & 2)
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1. Capstone Repository URL

· **Repository Name:*· `claimsense`
· **Local Git Branch:*· `main` (Commits: `3b4e914` [Milestone 1], `f23555f` [Mileston
e 2])
· **GitHub Repository URL:*· https://github.com/amirdhagopal/claimsense

2. Declaration of Completed Deliverables

Milestone 1: Discovery & Scoping (Week 1) · 100% COMPLETE

· [COMPLETED] **Engagement Charter (`docs/charter.md`):*· Complete 2-page document na
ming the three client systems (CMS, DocStore, PolicyAdmin), integration points, core constraints, and 5-week schedule.
· [COMPLETED] **Stakeholder Map & RACI Matrix (`docs/raci.md`):*· 9 identified stakeh
olders with influence·interest matrix, engagement-level RACI, and per-claim runtime RACI.
· [COMPLETED] **Four Quantified Success Criteria (`docs/success-criteria.md`):**
1. Decision accuracy (Target: ·90%, Minimum: ·80%)
2. Pipeline latency (Target: p50 ·30s, p95 ·60s)
3. Adjuster confirmation rate (Target: ·75%)
4. Fraud routing accuracy (Precision ·80%, Recall ·85%)
· [COMPLETED] **Architectural Decision Records (`DECISIONS.md`):*· ADRs 001 through 0
05 establishing the pre-agent deterministic spine, four-agent pipeline, hierarchical clause taxonomy, eval-driven development, and appen
· [COMPLETED] **Initial Evaluation Suite (`evals/cases/milestone1_cases.json`):*· 20
comprehensive evaluation cases covering all 4 outcomes (7 approve, 6 deny, 4 flag, 3 route) and 5 adversarial cases (prompt injection, cr
· [COMPLETED] **Evaluation Runner (`evals/run_evals.py`):*· Schema-validated runner v
erifying case distribution.

Milestone 2: Statement of Work, Pricing & Architecture (Week 2) · 100% COMPLETE

· [COMPLETED] **Statement of Work (`docs/sow.md`):*· 2-page formal contract covering
scope, acceptance criteria (AC-1 through AC-5), intellectual property, synthetic data governance, 5-week timeline, and $85,000 milestone
· [COMPLETED] **System Architecture & Data Flow (`docs/architecture.md`):*· Full Merm
aid architecture diagram with 16 labeled data flows, explicit deterministic spine vs. model judgment boundary justification, and 4 evaluated
· [COMPLETED] **Commercial Pricing & Unit Economics (`docs/pricing.md`):*· Staffing r
ates, AWS monthly infrastructure breakdown ($147.16/mo), per-agent OpenAI token consumption model ($0.040/claim), total cost ($0.10/
· [COMPLETED] **Risk Register (`docs/risk-register.md`):*· 6 detailed risks anticipat
ing later failures (prompt injection, cross-record leakage, stale policy data vs. lapsed dates, latency/quota limits, hallucinated citations, an
· [COMPLETED] **Typed Client System Contracts (`src/contracts/`, `docs/contracts.md`)
:*· Strongly-typed Pydantic v2 schemas for CMS, DocStore, and PolicyAdmin with UUIDv4 idempotency rules, SHA-256 payload fingerpri
· [COMPLETED] **Synthetic Data Shapes & Canonical Fixtures (`src/models/`, `data/fixt
ures/`):*· Motor policy clause catalog (`CLS-XXX-###`) and 20 validated synthetic claim dossiers.
· [COMPLETED] **Cloud Account Provisioning & Terraform IaC (`infra/`, `docs/cloud-pro
visioning.md`):*· Complete Terraform configuration for AWS ECS Fargate, ALB, ECR, CloudWatch Dashboard & Alarms, validated via `te

· [COMPLETED] **Architectural Decisions (`DECISIONS.md`):*· ADRs 006, 007, and 008 ad
ded for idempotency, Pydantic v2 contracts, and serverless container architecture.

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3. Environment Readiness & Execution Evidence

The project environment is verified and operational:

· **Python Runtime:*· Python 3.12+ with Pydantic v2, FastAPI, and Pytest.
· **Contract Verification:*· All contract unit tests pass with zero errors.
· **Evaluation Suite:*· All 20 evaluation cases pass schema validation and outcome di
stribution checks.
· **Terraform IaC:*· Initialized and validated with HashiCorp AWS Provider v5.100.0.

Execution Log Summary:

```
$ PYTHONPATH=. pytest tests/ -v
======================== 5 passed in 0.10s =========================
tests/test_contracts.py::test_claim_record_validation PASSED
tests/test_contracts.py::test_claim_record_invalid_id PASSED
tests/test_contracts.py::test_decision_request_idempotency_key_validation PASSED
tests/test_contracts.py::test_policy_deterministic_date_math PASSED
tests/test_contracts.py::test_clause_citation_regex PASSED
```

```
$ python3 evals/run_evals.py
============================================================
EVALUATION SUITE · CASE DISTRIBUTION
Total cases: 20
By decision outcome: approve: 7, deny: 6, flag: 4, route: 3
By category: standard: 15, adversarial: 5
Adversarial families: cross_record_leakage: 2, prompt_injection: 1,
                      forced_auto_decision: 1, narrative_contradiction: 1
============================================================
```

```
$ cd infra && terraform validate
Success! The configuration is valid.
```

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4. Statement of Outstanding Blockers

· **Technical / Design Blockers:*· **NONE.*· All architectural, scoping, contract, an
d evaluation deliverables for Milestones 1 and 2 are fully in place.
· **Account / Credential / Quota Status:**
- OpenAI API key and AWS environment credentials are ready for activation in Milestone 3 (Agent development).
- Terraform infrastructure templates are validated and ready to provision ECS Fargate and CloudWatch resources during deployment
- The stack is fully prepared for Milestone 3 (OpenAI Agents SDK implementation and CI evaluation gate).